

Probability for Data Science

Chapter 3: Discrete Random Variables and Probability Distributions

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Outline

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- Random Variables

- Probability Distributions for Discrete Random Variables

- The Cumulative Distribution Function

- Expected Values

- The Binomial Random Variable

- Hypergeometric and Negative Binomial Distributions

- The Poisson Probability Distribution

- The Poisson Process

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Random Variables

Definition

For a given sample space $\mathcal{S}$ of some experiment, a **random variable (rv)** is a function that associates a number with each outcome in $\mathcal{S}$. Hence, its domain is the sample space and its range is the set of real numbers.

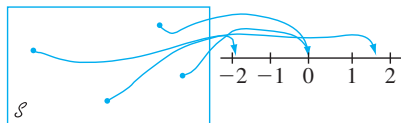


Figure: A random variable

Random variables are usually denoted by uppercase letters, such as X and Y . We will use lowercase letters to represent some particular value of the corresponding random variable. The notation $X(\omega) = x$ means that x is the value associated with the outcome $\omega \in \mathcal{S}$ by the rv X .

Random Variables

Example 1. When a student calls a university help desk for technical support, she/he will either immediately be able to speak to someone (S , for success) or will be placed on hold (F , for failure). With $\mathcal{S} = \{S, F\}$, we can define a rv X by

$$X(S) = 1 \quad \text{and} \quad X(F) = 0$$

The rv X indicates whether (1) or not (0) the student can immediately speak to someone. ■

Random Variables

Example 2. Consider the experiment in which a telephone number in a certain area code is dialed using a random number dialer (such devices are used extensively by polling organizations), and define a rv Y by

$$Y = \begin{cases} 1 & \text{if the selected number is unlisted} \\ 0 & \text{if the selected number is listed in the directory} \end{cases}$$

For example, if 5282966 appears in the telephone directory, then $Y(5282966) = 0$, whereas $Y(7727350) = 1$ tells us that the number 7727350 is unlisted.

A description of this sort is more economical than a complete listing, so we will use such a description whenever possible. ■

Random Variables

In the previous two examples, the only possible values of the random variable were 0 and 1. Such a random variable arises frequently enough to be given a special name, after the Swiss mathematician Jacob Bernoulli who first studied it.

Definition

Any random variable whose only possible values are 0 and 1 is called a **Bernoulli random variable**.

Random Variables

Example 3. Consider an experiment in which 9-volt batteries are tested until one with an acceptable voltage (S) is obtained. The sample space is $\mathcal{S} = \{S, FS, FFS, \dots\}$. Define a rv X by

$X =$ the number of batteries tested before the experiment terminates

Then $X(S) = 1$, $X(FS) = 2$, $X(FFS) = 3, \dots$, $X(FFFFFFFS) = 7$, and so on. Any positive integer is a possible value of X , so the set of possible values is infinite. ■

Random Variables

Example 4. Suppose that a location (latitude and longitude) in the continental United States is selected. Define a rv Y by

$Y =$ the height above sea level at the selected location

For example, if the selected location were $(39^\circ 50'N, 98^\circ 35'W)$, then we might have $Y((39^\circ 50'N, 98^\circ 35'W)) = 1748.26$ ft. The largest possible value of Y is 14494 (Mt. Whitney), and the smallest possible value is -282 (Death Valley). The set of all possible values of Y is the set of all numbers in the interval between -282 and 14494, that is

$$[-282, 14494] = \{y : y \text{ is a number, } -282 \leq y \leq 14494\}$$

and there are an infinite number of numbers in this interval. ■

Random Variables

Definition

A **discrete** random variable is a rv whose possible values either constitute a finite set or else can be listed in an infinite sequence in which there is a first element, a second element, and so on (“countably” infinite).

A random variable is **continuous** if both of the following apply:

1. Its set of possible values consists either of all numbers in a single interval on the number line (possibly infinite, e.g., from $-\infty$ to ∞) or all numbers in a disjoint union of such intervals (e.g., $[0, 10] \cup [20, 30]$).
2. No possible value of the variable has positive probability, that is, $P(X = c) = 0$ for any possible value c .

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Probability Distributions for Discrete Random Variables

The probability distribution of X says how the total probability of 1 is allocated to the various possible values of X . Suppose, for example, that a rv X takes the values 0, 1, 2, 3, and 4. The probability distribution will tell us how much probability is associated with the X value 0, how much is apportioned to the X value 1, and so on. We will use the following notation for the probabilities in the distribution:

$p(0)$ = the probability of the X value 0 = $P(X = 0)$

$p(1)$ = the probability of the X value 1 = $P(X = 1)$

and so on. In general, $p(x)$ will denote the probability assigned to the value x .

Probability Distributions for Discrete Random Variables

Example 5. Consider a lab with six computers reserved for statistics majors. Let X denote the number of these computers that are in use at a particular time of day. The probability distribution of a rv X is as given in the following table.

x	0	1	2	3	4	5	6
$p(x)$	.05	.10	.15	.25	.20	.15	.10

Compute:

1. The probability that at most 2 computers are in use.
2. The probability that at least 3 computers are in use.
3. The probability that between 2 and 5 computers inclusive are in use.
4. The probability that the number of computers in use is strictly between 2 and 5.

Probability Distributions for Discrete Random Variables

Example 5 (continued).

1. The probability that at most 2 computers are in use is

$$\begin{aligned}P(X \leq 2) &= P(X = 0 \text{ or } 1 \text{ or } 2) \\&= p(0) + p(1) + p(2) = .05 + .10 + .15 = .30\end{aligned}$$

2. Since the event *at least 3 computers are in use* is complementary to *at most 2 computers are in use*,

$$P(X \geq 3) = 1 - P(X \leq 2) = 1 - .30 = .70$$

which can, of course, also be obtained by adding together probabilities for the values 3, 4, 5, and 6.

Probability Distributions for Discrete Random Variables

Example 5 (continued).

3. The probability that between 2 and 5 computers inclusive are in use is

$$\begin{aligned}P(2 \leq X \leq 5) &= P(X = 2, 3, 4, \text{ or } 5) \\&= p(2) + p(3) + p(4) + p(5) \\&= .15 + .25 + .20 + .15 = .75\end{aligned}$$

4. The probability that the number of computers in use is strictly between 2 and 5 is

$$P(2 < X < 5) = P(X = 3 \text{ or } 4) = .25 + .20 = .45$$



Probability Distributions for Discrete Random Variables

Definition

The **probability distribution** or **probability mass function** (pmf) of a discrete rv is defined for every number x by

$$p(x) = P(X = x) = P(\text{all } \omega \in \mathcal{S} : X(\omega) = x)$$

Remark

The conditions

$$p(x) \geq 0 \quad \text{and} \quad \sum_{x \in \mathcal{S}} p(x) = 1$$

are required of any pmf.

Probability Distributions for Discrete Random Variables

Example 6. Six boxes of components are ready to be shipped by a certain supplier. The number of defective components in each box is as follows:

Box	1	2	3	4	5	6
Number of defectives	0	2	0	1	2	0

One of these boxes is to be randomly selected for shipment to a particular customer. Let X be the number of defectives in the selected box. Compute the pmf of X .

Probability Distributions for Discrete Random Variables

Example 6 (continued). The possible values of X are 0, 1, and 2. Of the six equally likely simple events, three result in $X = 0$, one in $X = 1$, and the other two in $X = 2$. Then

$$p(0) = P(X = 0) = P(\text{box 1 or 3 or 6 is sent}) = \frac{3}{6} = .5$$

$$p(1) = P(X = 1) = P(\text{box 4 is sent}) = \frac{1}{6} = .167$$

$$p(2) = P(X = 2) = P(\text{box 2 or 5 is sent}) = \frac{2}{6} = .333$$



Probability Distributions for Discrete Random Variables

Example 7. In a group of five potential blood donors $\{a, b, c, d, e\}$ only a and b have type O+ blood. Five blood samples, one from each individual, will be typed in random order until an O+ individual is identified. Let Y be the rv

Y = the number of typings necessary to identify an O+ individual

Compute the pmf of Y .

Probability Distributions for Discrete Random Variables

Example 7 (continued). The rv Y can take four values: 1, 2, 3 and 4 with the following probabilities

$$p(1) = P(Y = 1) = P(a \text{ or } b \text{ typed first}) = \frac{2}{5} = .4$$

$$\begin{aligned} p(2) &= P(Y = 2) = P(c, d, \text{ or } e \text{ first, and then } a \text{ or } b) \\ &= P(c, d, \text{ or } e \text{ first}) \cdot P(a \text{ or } b \text{ next} \mid c, d, \text{ or } e \text{ first}) \\ &= \frac{3}{5} \cdot \frac{2}{4} = .3 \end{aligned}$$

$$\begin{aligned} p(3) &= P(Y = 3) = P(c, d, \text{ or } e \text{ first and second, and then } a \text{ or } b) \\ &= \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{2}{3} = .2 \end{aligned}$$

$$\begin{aligned} p(4) &= P(Y = 4) = P(c, d, \text{ and } e \text{ all done first}) \\ &= \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{1}{3} = .1 \end{aligned}$$

Probability Distributions for Discrete Random Variables

Example 7 (continued). In tabular form, the pmf is

y	1	2	3	4
$p(y)$	.4	.3	.2	.1

We can represent the pmf on a line graph.

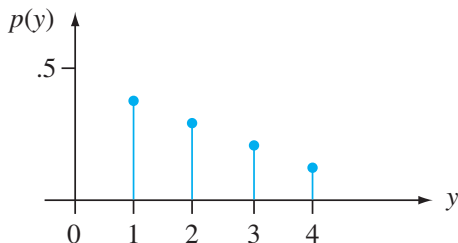


Figure: The line graph for the pmf

Probability Distributions for Discrete Random Variables

Example 7 (continued). We can also construct a **probability histogram**: Above each possible Y value y we draw a rectangle centered at y . The height of each rectangle is proportional to $p(y)$, and the base width is the same for all rectangles.

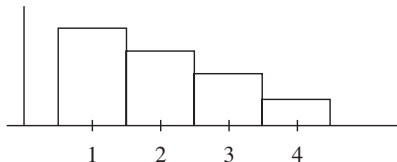


Figure: Probability histogram



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The Cumulative Distribution Function

For some fixed real number x , we often wish to compute the probability that the observed value of a rv X will be at most x .

Example 8. Let X be the number of beds occupied in a hospital's emergency room at a certain time of day; suppose the pmf of X is given by

x	0	1	2	3	4
$p(x)$	.20	.25	.30	.15	.10

Compute the followings: $P(X \leq 2)$, $P(X \leq 2.7)$, $P(X \leq -1.5)$, $P(X \leq 4)$, $P(X \leq 9.8)$ and $P(X < 2)$.

The Cumulative Distribution Function

Example 8 (continued).

- $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = p(0) + p(1) + p(2) = .75$
- Since $X \leq 2.7$ if and only if $X \leq 2$, we have $P(X \leq 2.7) = .75$
- $P(X \leq -1.5) = 0$ because 0 is the smallest possible value of X . We can also write $P(X \leq x) = 0$ for any negative number x .
- Since 4 is the largest possible value of X , we have $P(X \leq 4) = 1$ and $P(X \leq 9.8) = 1$. More generally, $P(X \leq x) = 1$ for any number $x \geq 4$.
- $P(X < 2) = P(X = 0) + P(X = 1) = p(0) + p(1) = .45$

Furthermore, $P(X \leq x)$ is a well-defined and computable probability for *any* number x . ■

The Cumulative Distribution Function

Definition

The **cumulative distribution function (cdf)** F of a discrete rv variable X with pmf p is defined for every number x by

$$F(x) = P(X \leq x) = \sum_{y: y \leq x} p(y)$$

For any number x , $F(x)$ is the probability that the observed value of X will be at most x .

The Cumulative Distribution Function

Example 9. A store carries flash drives with either 1 GB, 2 GB, 4 GB, 8 GB, or 16 GB of memory. The following table gives the distribution of

Y = the amount of memory in a purchased drive

y	1	2	4	8	16
$p(y)$	.05	.10	.35	.40	.10

Determine the cdf of Y .

The Cumulative Distribution Function

Example 9 (continued). Let's first determine $F(y)$ for each of the five possible values of Y :

- $F(1) = P(Y \leq 1) = P(Y = 1) = p(1) = .05$
- $F(2) = P(Y \leq 2) = P(Y = 1 \text{ or } 2) = p(1) + p(2) = .15$
- $F(4) = P(Y \leq 4) = P(Y = 1 \text{ or } 2 \text{ or } 4) = p(1) + p(2) + p(4) = .50$
- $F(8) = P(Y \leq 8) = p(1) + p(2) + p(4) + p(8) = .90$
- $F(16) = P(Y \leq 16) = 1$

The Cumulative Distribution Function

Example 9 (continued). Now for any other number y , $F(y)$ will equal the value of F at the closest possible value of Y to the left of y . For example,

$$F(2.7) = P(Y \leq 2.7) = P(Y \leq 2) = F(2) = .15$$

$$F(7.999) = P(Y \leq 7.999) = P(Y \leq 4) = F(4) = .50$$

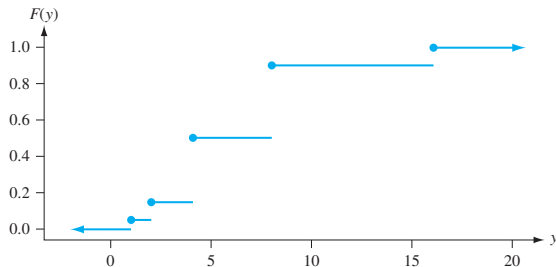
If y is less than 1, $F(y) = 0$ (e.g. $F(.58) = 0$), and if y is at least 16, $F(y) = 1$ (e.g., $F(25) = 1$). ■

The Cumulative Distribution Function

The cdf is thus

$$F(y) = \begin{cases} 0 & y < 1 \\ .05 & 1 \leq y < 2 \\ .15 & 2 \leq y < 4 \\ .50 & 4 \leq y < 8 \\ .90 & 8 \leq y < 16 \\ 1 & 16 \leq y \end{cases}$$

A graph of this cdf is shown in the following figure



The Cumulative Distribution Function

Proposition (Properties of a discrete rv's cdf)

The cdf F of a discrete rv X has the following properties:

- The range of F is $[0, 1]$.
- F is **right continuous**.
- F is a **non-decreasing step function**. In other words, the graph of F will have a jump at every possible value of X and will be flat between possible values, with steps going up.
- The amplitude of the jump at x equals $P(X = x)$.
- $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.

The Cumulative Distribution Function

Example 10. Starting at a fixed time, we observe the gender of each newborn child at a certain hospital until a boy (B) is born. Let $p = P(B)$, assume that successive births are independent, and define the rv X by

X = the number of births up to and including that of the first boy

Then

$$p(1) = P(X = 1) = P(B) = p$$

$$p(2) = P(X = 2) = P(GB) = P(G) \cdot P(B) = (1 - p)p$$

$$p(3) = P(X = 3) = P(GGB) = P(G) \cdot P(G) \cdot P(B) = (1 - p)^2 p$$

Continuing in this way, a general formula emerges:

$$p(x) = (1 - p)^{x-1} p \quad \text{for } x = 1, 2, 3, \dots$$

Such a rv is called a **geometric random variable with parameter p** . The parameter p can assume any value between 0 and 1.

The Cumulative Distribution Function

Example 10 (continued). For any positive integer x ,

$$\begin{aligned} F(x) &= \sum_{y \leq x} p(y) = \sum_{y=1}^x (1-p)^{y-1} p \\ &= p \sum_{y=0}^{x-1} (1-p)^y = 1 - (1-p)^x \end{aligned}$$

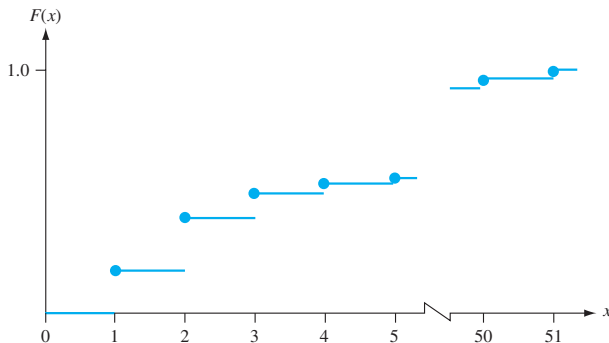
Since F is constant in between positive integers,

$$F(x) = \begin{cases} 0 & x < 1 \\ 1 - (1-p)^{\lfloor x \rfloor} & x \geq 1 \end{cases}$$

where $\lfloor x \rfloor$ is the largest integer $\leq x$.

The Cumulative Distribution Function

Example 10 (continued). If $p = .51$, then the cdf of X is graphed in the following figure.



Clearly, F is a non-decreasing and right continuous step function. ■

The Cumulative Distribution Function

So far, the cdf has been derived from the pmf. This process can be reversed to obtain the pmf from the cdf whenever the latter function is available.

Example 11. Consider again the rv of Example 9:

y	1	2	4	8	16
$p(y)$	.05	.10	.35	.40	.10

We can write

$$\begin{aligned} p(8) &= P(Y = 8) \\ &= [p(1) + p(2) + p(4) + p(8)] - [p(1) + p(2) + p(4)] \\ &= P(Y \leq 8) - P(Y \leq 4) \\ &= F(8) - F(4) \end{aligned}$$

The Cumulative Distribution Function

Example 11 (continued). More generally, the probability that Y falls in a specified interval is easily obtained from the cdf. For example,

$$\begin{aligned}P(4 \leq Y \leq 8) &= p(4) + p(8) \\&= [p(1) + p(2) + p(4) + p(8)] - [p(1) + p(2)] \\&= P(Y \leq 8) - P(Y \leq 2) \\&= F(8) - F(2)\end{aligned}$$

Notice that $P(4 \leq Y \leq 8) \neq F(8) - F(4)$. This is because the Y value 4 is included in $4 \leq Y \leq 8$, so we do not want to subtract out its probability. However, $P(2 < Y \leq 8) = F(8) - F(2)$ because $Y = 2$ is not in the interval $2 < Y \leq 8$. ■

The Cumulative Distribution Function

Proposition

For any two numbers a and b with $a \leq b$,

$$P(a \leq X \leq b) = F(b) - F(a^-)$$

$$P(a < X \leq b) = F(b) - F(a)$$

$$P(a < X < b) = F(b^-) - F(a)$$

$$P(a \leq X < b) = F(b^-) - F(a^-)$$

where “ a^- ” (resp. “ b^- ”) represents the largest possible value of X that is strictly less than a (resp. b). In particular, if the only possible values are integers and if a and b are integers, then $a^- = a - 1$ and $b^- = b - 1$.

Taking $a = b$ yields $P(X = a) = F(a) - F(a^-)$.

The Cumulative Distribution Function

Example 12. Let X denote the number of days of sick leave taken by a randomly selected employee of a large company during a particular year. If the maximum number of allowable sick days per year is 14, possible values of X are $0, 1, \dots, 14$. Knowing that $F(0) = .58$, $F(1) = .72$, $F(2) = .76$, $F(3) = .81$, $F(4) = .88$, and $F(5) = .94$, compute $P(2 \leq X \leq 5)$ and $P(X = 3)$.

The Cumulative Distribution Function

Example 12 (continued). Using the cdf, we get

$$P(2 \leq X \leq 5) = F(5) - F(2^-) = F(5) - F(1) = .94 - .72 = .22$$

and

$$P(X = 3) = F(3) - F(3^-) = F(3) - F(2) = .81 - .76 = .05$$



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Example 13. Consider a university having 15000 students and let X denote the number of courses for which a randomly selected student is registered.

x	1	2	3	4	5	6	7
Number registered	150	450	1950	3750	5850	2550	300

The average number of courses per student, or the average value of X in the population is

$$\frac{1(150) + 2(450) + 3(1950) + 4(3750) + 5(5850) + 6(2550) + 7(300)}{15000} = 4.57$$

Expected Values

Example 13 (continued). An alternative consists of computing the pmf of X

$$\begin{aligned} p(1) &= \frac{150}{15000} = .01 & p(2) &= \frac{450}{15000} = .03 & p(3) &= \frac{1950}{15000} = .13 \\ p(4) &= \frac{3750}{15000} = .25 & p(5) &= \frac{5850}{15000} = .39 & p(6) &= \frac{2550}{15000} = .17 \\ p(7) &= \frac{300}{15000} = .02 \end{aligned}$$

and to take the population average value of X as

$$\begin{aligned} 1 \cdot p(1) + 2 \cdot p(2) + 3 \cdot p(3) + 4 \cdot p(4) + 5 \cdot p(5) + 6 \cdot p(6) + 7 \cdot p(7) \\ = 4.57 \end{aligned}$$



Expected Values

Definition

Let X be a discrete rv with set of possible values D and pmf $p(x)$. The **expected value** or **mean value** of X , denoted by $E(X)$ or μ_X or just μ , is

$$E(X) = \sum_{x \in D} x \cdot p(x)$$

We also refer to μ as the **population mean** rather than the mean of X in the population.

In words, the mean value of X is the *weighted average* of the possible X values where the weights are the probabilities of those values.

Expected Values

Example 14. Consider the rv X with the following pmf

x	1	2	3	4	5	6	7
$p(x)$	.01	.03	.13	.25	.39	.17	.02

The expected value of X is

$$\mu = (1) \cdot p(1) + \cdots + (7) \cdot p(7) = 4.57$$

Notice that:

1. μ here is not the ordinary average of 1, 2, 3, 4, 5, 6, 7 which is 4 because the distribution puts more weight on 4, 5 and 6 than on other X values.
2. μ here is not a possible value of X .



Expected Values

Example 15. Let $X = 1$ if a randomly selected vehicle passes an emissions test and $X = 0$ otherwise. Then X is a Bernoulli rv with pmf $p(1) = p$ and $p(0) = 1 - p$, from which

$$E(X) = 0 \cdot p(0) + 1 \cdot p(1) = p$$

That is, the expected value of X is just the probability that X takes on the value 1. 

Expected Values

Example 16. Let X be the number of children born up to and including the first boy. We know that

$$p(x) = p(1 - p)^{x-1} \quad x = 1, 2, 3, \dots$$

The expectation of X is

$$\begin{aligned} E(X) &= \sum_{x=1}^{\infty} x \cdot p(x) = \sum_{x=1}^{\infty} x \cdot p(1 - p)^{x-1} = p \sum_{x=1}^{\infty} \left[-\frac{d}{dp} (1 - p)^x \right] \\ &= -p \frac{d}{dp} \left[\sum_{x=1}^{\infty} (1 - p)^x \right] = -p \frac{d}{dp} \left[\frac{1 - p}{1 - (1 - p)} \right] = -p \frac{d}{dp} \left[\frac{1 - p}{p} \right] = \frac{1}{p} \end{aligned}$$

If p is near 1, we expect to see a boy very soon, whereas if p is near 0, we expect many births before the first boy. For $p = .5$, $E(X) = 2$. ■

Expected Values

Example 17. Consider a rv X with the following pmf

x	4	6	8
$p(x)$	.5	.3	.2

Compute the expected value of the rv Y defined by:

$$Y = h(X) = 20 + 3X + .5X^2$$

Expected Values

Example 17 (continued). The derived pmf of Y is

y	40	56	76
$p(y)$	.5	.3	.2

Thus, if D^* denotes the possible values of Y ,

$$\begin{aligned} E(Y) &= E[h(X)] = \sum_{y \in D^*} y \cdot p(y) \\ &= (40) \cdot (.5) + (56) \cdot (.3) + (76) \cdot (.2) = 52 \end{aligned}$$

Notice that

$$\begin{aligned} E(Y) &= (40) \cdot (.5) + (56) \cdot (.3) + (76) \cdot (.2) \\ &= h(4) \cdot p(4) + h(6) \cdot p(6) + h(8) \cdot p(8) \\ &= \sum_{x \in D} h(x) \cdot p(x) \end{aligned}$$

Hence, $E[h(X)]$ is computed in the same way that $E(X)$ itself is, except that $h(x)$ is substituted in place of x .

Expected Values

Proposition

If the rv X has a set of possible values D and pmf $p(x)$, then the expected value of any function $h(X)$, denoted by $E[h(X)]$ or $\mu_{h(X)}$, is computed by

$$E[h(X)] = \sum_{x \in D} h(x) \cdot p(x)$$

Expected Values

Example 18. A computer store has purchased three computers of a certain type at \$500 apiece. It will sell them for \$1000 apiece. The manufacturer has agreed to repurchase any computers still unsold after a specified period at \$200 apiece. Let X denote the number of computers sold, and suppose that

$$p(0) = .1, \quad p(1) = .2, \quad p(2) = .3, \quad \text{and} \quad p(3) = .4$$

Compute the expected profit associated with selling X units.

Expected Values

Example 18 (continued). Let $h(X)$ denote the profit associated with selling X units, the given information implies that

$$h(X) = \text{revenue} - \text{cost} = 1000X + 200(3 - X) - 1500 = 800X - 900$$

This leads to

$$\begin{aligned} E[h(X)] &= E(800X - 900) \\ &= (-900) \cdot p(0) + (800 - 900) \cdot p(1) + (1600 - 900) \cdot p(2) \\ &\quad + (2400 - 900) \cdot p(3) \\ &= (-900) \cdot (.1) + (-100) \cdot (.2) + (700) \cdot (.3) + (1500) \cdot (.4) \\ &= 700 \end{aligned}$$

Thus, the expected profit is then \$700. ■

Expected Values

When $h(X)$ is an affine function $aX + b$, $E[h(X)]$ is easily computed from $E(X)$.

Proposition

$$E(aX + b) = aE(X) + b$$

Or, using alternative notation, $\mu_{aX+b} = a\mu_X + b$.

In fact,

$$\begin{aligned} E(aX + b) &= \sum_{x \in D} (ax + b) \cdot p(x) \\ &= a \sum_{x \in D} x \cdot p(x) + b \sum_{x \in D} p(x) = aE(X) + b \end{aligned}$$

Remark

1. For any constant a , $E(aX) = aE(X)$.
2. For any constant b , $E(X + b) = E(X) + b$.

The Variance

We use the variance of X to assess the amount of variability in (the distribution of) X , just as s^2 was used previously to measure variability in a sample.

Definition

Let X have pmf $p(x)$ and expected value μ . Then the **variance** of X , denoted by $V(X)$ or σ_X^2 , or just σ^2 , is

$$V(X) = E[(X - \mu)^2] = \sum_{x \in D} (x - \mu)^2 \cdot p(x)$$

The **standard deviation** (SD) of X is

$$\sigma_X = \sqrt{\sigma_X^2}$$

The Variance

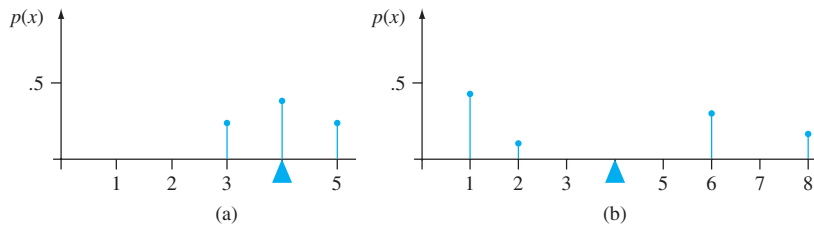


Figure: Two different probability distributions with the same mean $\mu = 4$. The distribution in (b) has greater spread (i.e., variability or dispersion) than does that in (a).

The Variance

Proposition (A Shortcut Formula for σ^2)

$$V(X) = \left(\sum_{x \in D} x^2 \cdot p(x) \right) - \mu^2 = E(X^2) - [E(X)]^2$$

The proof is straightforward. It consists of expanding $(x - \mu)^2$ in the definition of σ^2 to obtain $x^2 - 2x\mu + \mu^2$, and then carry $\sum$ through to each of the three terms.

The Variance

Example 19. Consider a rv X having the following pmf:

x	1	2	3	4	5	6
$p(x)$	.30	.25	.15	.05	.10	.15

The expected value of X is

$$\begin{aligned}\mu &= (1) \cdot (.30) + (2) \cdot (.25) + (3) \cdot (.15) + (4) \cdot (.05) \\ &\quad + (5) \cdot (.10) + (6) \cdot (.15) = 2.85\end{aligned}$$

The variance of X is then

$$\begin{aligned}V(X) &= \sum_{i=1}^6 (x - 2.85)^2 \cdot p(x) \\ &= (1 - 2.85)^2 \cdot (.30) + (2 - 2.85)^2 \cdot (.25) + (3 - 2.85)^2 \cdot (.15) \\ &\quad + (4 - 2.85)^2 \cdot (.05) + (5 - 2.85)^2 \cdot (.10) + (6 - 2.85)^2 \cdot (.15) \\ &= 3.2275\end{aligned}$$

The standard deviation of X is $\sigma = \sqrt{3.2275} = 1.7965$.

The Variance

Example 19 (continued). Using the shortcut formula for σ^2 gives

$$\begin{aligned} E(X^2) &= \sum_{i=1}^6 x^2 \cdot p(x) \\ &= (1)^2 \cdot (.30) + (2)^2 \cdot (.25) + (3)^2 \cdot (.15) + (4)^2 \cdot (.05) \\ &\quad + (5)^2 \cdot (.10) + (6)^2 \cdot (.15) \\ &= 11.35 \end{aligned}$$

Thus,

$$V(X) = E(X^2) - [E(X)]^2 = 11.35 - (2.85)^2 = 3.2275$$



The Variance

The variance of $h(X)$ is the expected value of the squared difference between $h(X)$ and its expected value:

$$V[h(X)] = \sum_{x \in D} \{h(X) - E[h(X)]\}^2 \cdot p(x)$$

When $h(X) = aX + b$ is an affine function, arithmetic operations give a simple relationship between $V[h(X)]$ and $V(X)$.

Proposition

$$V(aX + b) = a^2 V(X) \quad \text{and} \quad \sigma_{aX+b} = |a| \cdot \sigma_X$$

In particular,

$$\sigma_{aX} = |a| \cdot \sigma_X \quad \text{and} \quad \sigma_{X+b} = \sigma_X$$

The Variance

Example 20. In the computer sales scenario of Example 18,

$$E(X) = (0) \cdot (.1) + (1) \cdot (.2) + (2) \cdot (.3) + (3) \cdot (.4) = 2$$

and

$$E(X^2) = (0)^2 \cdot (.1) + (1)^2 \cdot (.2) + (2)^2 \cdot (.3) + (3)^2 \cdot (.4) = 5$$

so $V(X) = 5 - (2)^2 = 1$.

The profit function $h(X) = 800X - 900$ then has variance

$$V[h(X)] = V(800X - 900) = (800)^2 V(X) = (640000) \cdot (1) = 640000$$

and standard deviation 800. ■

High-order Moments

Definition

The **moment of order k** or the **k^{th} moment** of X is

$$E(X^k) = \sum_{x \in D} x^k \cdot p(x)$$

The **central moment of order k** or the **k^{th} central moment** of X is

$$E[(X - \mu)^k]$$

Thus, the expected value of X is its moment of order 1. The variance of X is its central moment of order 2.

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The Binomial Random Variable

Definition

A **binomial experiment** is a random experiment that conforms the following requirements:

1. The experiment consists of a sequence of n smaller experiments called trials, where n is fixed in advance of the experiment.
2. Each trial can result in one of the same two possible outcomes, which we generically denote by success (S) and failure (F).
3. The trials are independent, so that the outcome on any particular trial does not influence the outcome on any other trial.
4. The probability of success $P(S)$ is constant from trial to trial; we denote this probability by p .

Hence, a binomial experiment is a replication of n trials that are dichotomous, independent, and homogenous.

The Binomial Random Variable

Example 21.

1. Consider each of n vehicles undergoing an emissions test to track the pollutants they release, and let S denote a vehicle that passes the test and F denote one that fails to pass. Then this experiment is binomial.
2. Tossing a coin n times, with S for getting Heads and F for getting Tails, also results in a binomial experiment.
3. The experiment in which the gender (S for female and F for male) is determined for each of the next n children born at a particular hospital is binomial.
4. Consider the experiment of rolling a die ten times. Each new outcome is a success if the number that shows on the top face is 5. With this identification of S and F , the experiment is binomial with $n = 10$ and $p = P(S) = \frac{1}{6}$.



The Binomial Random Variable

Example 22. A bag contains 5 red balls and 5 black balls. Suppose that 3 of these balls are randomly selected and label the i th trial as a success S if the drawn ball is red and a failure F otherwise. If the ball is replaced in the bag after each selection, then the probability p of a success stays constant (we always have a probability of .5 of selecting a red ball next).

However, if the ball are drawn one by one (without replacement), then this affects the probability for future selections. In fact,

$$P(S \text{ on } 2^{\text{nd}} \text{ trial} | S \text{ on } 1^{\text{st}} \text{ trial}) = \frac{4}{9}$$

and

$$P(S \text{ on } 3^{\text{rd}} \text{ trial} | S \text{ on } 1^{\text{st}} \text{ and } 2^{\text{nd}} \text{ trials}) = \frac{3}{8}$$

and so forth.

The Binomial Random Variable

Example 22 (continued). Similarly,

$$P(S \text{ on } 2^{\text{nd}} \text{ trial} | F \text{ on } 1^{\text{st}} \text{ trial}) = \frac{5}{9}$$

and

$$P(S \text{ on } 3^{\text{rd}} \text{ trial} | F \text{ on } 1^{\text{st}} \text{ and } 2^{\text{nd}} \text{ trials}) = \frac{5}{8}$$

and so forth.

This experiment is *not* binomial because the trials are not independent. In general, *if sampling is without replacement, the experiment will not yield independent trials.* ■

The Binomial Random Variable

Example 23. Consider a large country that has 500,000 individuals in its jury pool, of whom 400,000 are employed. A sample of 10 individuals from the pool is chosen without replacement. The i^{th} trial is regarded as a success S if the i^{th} individual is employed. Notice that the size of the population being sampled is very large relative to the sample size $n = 10$. In this case

$$P(S \text{ on } 2^{\text{nd}} \text{ trial} | S \text{ on } 1^{\text{st}} \text{ trial}) = \frac{399,999}{499,999} = .799999 \approx .80$$

and

$$P(S \text{ on } 10^{\text{th}} \text{ trial} | S \text{ on first 9 trials}) = \frac{399,991}{499,991} = .799996 \approx .80$$

$$P(S \text{ on } 10^{\text{th}} \text{ trial} | F \text{ on first 9 trials}) = \frac{400,000}{499,991} = .800014 \approx .80$$

The Binomial Random Variable

Example 23 (continued). These calculations suggest that although the trials are not exactly independent, the conditional probabilities differ so slightly from one another that for practical purposes the trials can be regarded as independent with constant $P(S) = .8$. Thus, to a very good approximation, the experiment is binomial with $n = 10$ and $p = .8$. ■

The Binomial Random Variable

We will use the following rule of thumb in deciding whether a “without replacement” experiment can be treated as being binomial.

Rule

Consider sampling without replacement from a dichotomous population of size N . If the sample size (number of trials) n is at most 5% of the population size (i.e. $n/N \leq .05$ or equivalently, $N \geq 20n$), the experiment can be analyzed as though it were a binomial experiment.

In Example 22, $n/N = 3/10 = .3 > .05$, so the binomial experiment is not a good approximation, but in Example 23, $n/N = 10/500,000 \ll .05$, so we can use the binomial approximation.

The Hypergeometric Distribution

Example 24. Consider selecting $n = 3$ children from a city where N children live and suppose that 50% of the children are girls and 50% are boys. The following table shows the probabilities of selecting 3 boys when sampling with or without replacement and for different values of N .

N	With replacement	Without replacement	n/N
20	$\left(\frac{10}{20}\right)^3 = .125$	$\frac{10}{20} \cdot \frac{9}{19} \cdot \frac{8}{18} \approx .105$	.15
30	$\left(\frac{15}{30}\right)^3 = .125$	$\frac{15}{30} \cdot \frac{14}{29} \cdot \frac{13}{28} \approx .112$	.10
100	$\left(\frac{50}{100}\right)^3 = .125$	$\frac{50}{100} \cdot \frac{49}{99} \cdot \frac{48}{98} \approx .121$	.03 < .05
10000	$\left(\frac{5000}{10000}\right)^3 = .125$	$\frac{5000}{10000} \cdot \frac{4999}{9999} \cdot \frac{4998}{9998} \approx .125$	.0003 < .05



The Binomial Random Variable

Definition

The binomial random variable X associated with a binomial experiment consisting of n trials is defined as

X = the number of S 's among the n trials

The Binomial Random Variable

Suppose, for example, that $n = 3$. Then there are eight possible outcomes for the experiment:

$$SSS \quad SSF \quad SFS \quad SFF \quad FSS \quad FSF \quad FFS \quad FFF$$

From the definition of X ,

$$X(SSS) = 3 \quad X(SSF) = 2 \quad X(SFS) = 2 \quad X(SFF) = 1$$

$$X(FSS) = 2 \quad X(FSF) = 1 \quad X(FFS) = 1 \quad X(FFF) = 0$$

Consequently, possible values for X in an n -trial experiment are $x = 0, 1, 2, \dots, n$. We will often write $X \sim \text{Bin}(n, p)$ to indicate that X is a binomial rv based on n trials with success probability p .

Because the pmf of a binomial rv X depends on the two parameters n and p , we denote the pmf by $b(x; n, p)$.

The Binomial Random Variable

Consider a binomial experiment consisting of $n = 4$ trials with some success probability p ($0 < p < 1$) and denote by X the corresponding rv, so $X \sim \text{Bin}(4, p)$ for some $0 < p < 1$. The outcomes of the experiment, their probabilities and the corresponding X values are displayed in the following table:

Outcome	x	Probability	Outcome	x	Probability
SSSS	4	p^4	FSSS	3	$p^3(1 - p)$
SSSF	3	$p^3(1 - p)$	FSSF	2	$p^2(1 - p)^2$
SSFS	3	$p^3(1 - p)$	FSFS	2	$p^2(1 - p)^2$
SSFF	2	$p^2(1 - p)^2$	FSFF	1	$p(1 - p)^3$
SFSS	3	$p^3(1 - p)$	FFSS	2	$p^2(1 - p)^2$
SFSF	2	$p^2(1 - p)^2$	FFSF	1	$p(1 - p)^3$
SFFS	2	$p^2(1 - p)^2$	FFFS	1	$p(1 - p)^3$
SFFF	1	$p(1 - p)^3$	FFFF	0	$(1 - p)^4$

The Binomial Random Variable

We want to compute $b(x; 4, p)$ for $x = 0, 1, 2, 3$, and 4.

For $b(3; 4, p)$, we identify which of the 16 outcomes yield an x value of 3 and sum the probabilities associated with each such outcome:

$$b(3; 4, p) = P(FSSS) + P(SFSS) + P(SSFS) + P(SSSF) = 4p^3(1-p)$$

There are four outcomes with $X = 3$ and each has probability $p^3(1-p)$ (the order of S 's and F 's is not important, only the number of S 's), so

$$b(3; 4, p) = \{\text{number of outcomes with } X = 3\} \\ \cdot \{\text{probability of any particular outcome with } X = 3\}$$

The Binomial Random Variable

In general,

$$b(x; n, p) = \{\text{number of sequences of length } n \text{ consisting of } x \text{ } S\text{'s}\} \\ \cdot \{\text{probability of any particular such sequence}\}$$

Since the ordering of S 's and F 's is not important, the second factor in the previous equation is $p^x(1-p)^{n-x}$. The first factor is the number of words of length n that can be written using x S 's and $n-x$ F 's. Hence, it is $\binom{n}{x}$.

Theorem

$$b(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x} \quad \text{for } x = 0, 1, 2, \dots, n$$

The Binomial Random Variable

Example 25. Each of six randomly selected cola drinkers is given a glass containing cola S and one containing cola F . The glasses are identical in appearance and suppose there is actually no tendency among cola drinkers to prefer one cola to the other. Let

X = the number among the six who prefer S

Compute:

1. The probability that three prefer S .
2. The probability that at least three prefer S .
3. The probability that at most one prefers S .
4. The probability that no one prefers F .

The Binomial Random Variable

Example 25 (continued).

Let $p = P(\text{a selected individual prefers } S)$. Since there is no tendency among cola drinkers to prefer one cola to the other, we can set $p = .5$. If

$X = \text{the number among the six who prefer } S$

then $X \sim \text{Bin}(6, .5)$. Thus

1. The probability that three prefer S is

$$P(X = 3) = b(3; 6, .5) = \binom{6}{3} (.5)^3 (.5)^3 = .3125$$

2. The probability that at least three prefer S is

$$P(3 \leq X) = \sum_{x=3}^6 b(x; 6, .5) = \sum_{x=3}^6 \binom{6}{x} (.5)^x (.5)^{6-x} = .6563$$

The Binomial Random Variable

Example 25 (continued).

3. The probability that at most one prefers S is

$$P(X \leq 1) = \sum_{x=0}^1 b(x; 6, .5) = .1094$$

4. The probability that no one prefers F is

$$P(X = 6) = \binom{6}{6} (.5)^6 (.5)^0 = .0156$$



The Binomial Random Variable

Proposition

For $X \sim \text{Bin}(n, p)$, the cdf will be denoted by $B(x; n, p)$. For any number x , we have

$$B(x; n, p) = P(X \leq x) = \begin{cases} \sum_{y=0}^{\lfloor x \rfloor} b(y; n, p) & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Appendix Table A.1 in the Textbook tabulates the cdf $B(x; n, p)$ for some values of n and p .

The Binomial Random Variable

Proposition

If $X \sim \text{Bin}(n, p)$, then

$$E(X) = np, \quad V(X) = np(1 - p) = npq, \quad \text{and} \quad s_X = \sqrt{npq}$$

where $q = 1 - p$.

The Binomial Random Variable

Proof. The definition of the expected value gives

$$\begin{aligned} E(X) &= \sum_{x=0}^n x \cdot p(x) = \sum_{x=1}^n x \binom{n}{x} p^x q^{n-x} = \sum_{x=1}^n x \frac{n!}{x!(n-x)!} p^x q^{n-x} \\ &= \sum_{x=1}^n x \frac{(n-1)!n}{(x-1)!x(n-x)!} p^{x-1} p q^{n-x} \\ &= np \sum_{x=1}^n \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} q^{n-x} \\ &= np \sum_{y=0}^{n-1} \frac{(n-1)!}{y!(n-1-y)!} p^y q^{(n-1)-y} \\ &= np \sum_{y=0}^{n-1} \binom{n-1}{y} p^y q^{(n-1)-y} \end{aligned}$$

Notice that the last sum is that of the pmf of a binomial rv $\text{Bin}(n-1, p)$; so it is equal to 1. Thus, $E(X) = np$.

The Binomial Random Variable

Proof (continued). To compute the variance of X , we start calculating $E(X^2)$.

$$\begin{aligned} E(X^2) &= \sum_{x=0}^n x^2 \cdot p(x) = \sum_{x=1}^n x^2 \binom{n}{x} p^x q^{n-x} \\ &= \sum_{x=1}^n x(x-1+1) \binom{n}{x} p^x q^{n-x} \\ &= \sum_{x=1}^n x(x-1) \binom{n}{x} p^x q^{n-x} + \sum_{x=1}^n \binom{n}{x} p^x q^{n-x} \\ &= \sum_{x=2}^n x(x-1) \binom{n}{x} p^x q^{n-x} + np \\ &= \sum_{x=2}^n x(x-1) \frac{n!}{x!(n-x)!} p^x q^{n-x} + np \\ &= \sum_{x=2}^n x(x-1) \frac{(n-2)!(n-1)n}{(x-2)!(x-1)x(n-x)!} p^{x-2} p^2 q^{n-x} + np \end{aligned}$$

The Binomial Random Variable

Proof (continued). Proceeding as in the computation of $E(X)$, gives

$$\begin{aligned} E(X^2) &= \sum_{x=2}^n \frac{(n-2)!(n-1)n}{(x-2)!(n-x)!} p^{x-2} p^2 q^{n-x} + np \\ &= (n-1)np^2 \sum_{x=2}^n \frac{(n-2)!}{(x-2)!(n-x)!} p^{x-2} q^{n-x} + np \\ &= (n-1)np^2 \sum_{y=0}^{n-2} \frac{(n-2)!}{y!(n-2-y)!} p^y q^{(n-2)-y} + np \\ &= (n-1)np^2 \underbrace{\sum_{y=0}^{n-2} \binom{n-2}{y} p^y q^{(n-2)-y}}_{=1} + np \\ &= (n-1)np^2 + np \end{aligned}$$

The Binomial Random Variable

Proof (continued). Therefore, the variance of X is

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 \\ &= (n-1)np^2 + np - (np)^2 \\ &= n^2p^2 - np^2 + np - n^2p^2 \\ &= np - np^2 \\ &= np(1-p) \\ &= npq \end{aligned}$$



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The Hypergeometric Distribution

The assumptions leading to the hypergeometric distribution are as follows:

1. The population or set to be sampled consists of N individuals, objects, or elements (a finite population).
2. Each individual can be characterized as a success (S) or a failure (F), and there are M successes in the population.
3. A sample of n individuals is selected without replacement in such a way that each subset of size n is equally likely to be chosen.

The random variable of interest is

$X =$ the number of S 's in the sample

The probability distribution of X depends on the parameters n , M , and N , so we wish to obtain $P(X = x) = h(x; n, M, N)$.

The Hypergeometric Distribution

Example 26. During a particular period a university's IT office received 20 service orders for problems with printers, of which 8 were laser printers and 12 were inkjet models. A sample of 5 of these service orders is to be selected for inclusion in a customer satisfaction survey. Suppose that the 5 are selected in a completely random fashion, so that any particular subset of size 5 has the same chance of being selected as does any other subset. What is the probability that exactly two of the selected service orders were for inkjet printers?

The Hypergeometric Distribution

Example 26 (continued). Here, the population size is $N = 20$, the sample size is $n = 5$, and the number of S 's (inkjet= S) and F 's in the population are $M = 12$ and $N - M = 8$, respectively. Let

X = the number of service orders that are for inkjet printers

Because all outcomes (each consisting of 5 particular orders) are equally likely,

$$P(X = 2) = h(2; 5, 12, 20) = \frac{\binom{12}{2}\binom{8}{3}}{\binom{20}{5}} = \frac{3696}{15504} = .238$$

Similarly, we can compute $P(X = x)$ for $x = 0, 1, 3, 4$, and 5. ■

The Hypergeometric Distribution

Remark

In general, if the sample size n is smaller than the number of successes M in the population, then the largest possible X value is n . However, if $M < n$, then X can be at most M .

Similarly, whenever the number of population failures $N - M$ exceeds the sample size n , the smallest possible X value is 0 (since all sampled individuals might then be failures).

However, if $N - M < n$, the smallest possible X value is $n - (N - M)$.

Thus, the possible values of X satisfy the restriction

$$\max(0, n - (N - M)) \leq x \leq \min(n, M)$$

The Hypergeometric Distribution

Proposition

If X is the number of S 's in a completely random sample of size n drawn from a population consisting of M S 's and $(N - M)$ F 's, then the probability distribution of X , called the **hypergeometric distribution**, is given by

$$P(X = x) = h(x; n, M, N) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$$

for an integer x satisfying

$$\max(0, n - (N - M)) \leq x \leq \min(n, M)$$

The Hypergeometric Distribution

Proposition

The mean and variance of the hypergeometric rv X having pmf $h(x; n, M, N)$ are

$$E(X) = n \cdot \frac{M}{N} \quad \text{and} \quad V(X) = \left(\frac{N-n}{N-1} \right) \cdot n \cdot \frac{M}{N} \cdot \left(1 - \frac{M}{N} \right)$$

The ratio M/N is the proportion of S 's in the population. Replacing M/N by p in $E(X)$ and $V(X)$ gives

$$E(X) = np \quad \text{and} \quad V(X) = \left(\frac{N-n}{N-1} \right) \cdot np(1-p)$$

The Hypergeometric Distribution

Remark

Our general rule of thumb stated that when sampling without replacement but n/N was at most .05, then the binomial distribution could be used to compute approximate probabilities involving the number of S 's in the sample.

The rationale for the rule is the following: Let the population size, N , and number of population S 's, M , get large with the ratio M/N approaching p . Then $h(x; n, M, N)$ approaches $b(x; n, p)$; so for n/N small, the two are approximately equal provided that p is not too near either 0 or 1.

The Negative Binomial Distribution

Definition

The negative binomial rv and distribution are based on an experiment satisfying the following conditions:

1. The experiment consists of a sequence of independent trials.
2. Each trial can result in either a success (S) or a failure (F).
3. The probability of success is constant from trial to trial, so $P(S \text{ on trial } i) = p$ for $i = 1, 2, 3, \dots$
4. The experiment continues (trials are performed) until a total of r successes have been observed, where r is a specified positive integer.

The random variable of interest is

X = the number of failures that precede the r^{th} success

X is called a **negative binomial random variable** because, in contrast to the binomial rv, the number of successes is fixed and the number of trials is random.

The Negative Binomial Distribution

Possible values of a negative binomial rv X are $0, 1, 2, \dots$

We denote by $nb(x; r, p)$ the pmf of X .

Consider $nb(7; 3, p) = P(X = 7)$, the probability that exactly 7 F 's occur before the third S . In order for this to happen, the 10th trial must be an S and there must be exactly 2 S 's and 7 F 's among the first 9 trials. Thus

$$nb(7; 3, p) = \binom{9}{7} (1-p)^7 p^3$$

The Negative Binomial Distribution

Proposition

The pmf of the negative binomial rv X with parameters $r = \text{number of } S\text{'s}$ and $p = P(S)$ is

$$nb(x; r, p) = \binom{x + r - 1}{r - 1} (1 - p)^x p^r \quad x = 0, 1, 2, \dots$$

Proposition

If X is a negative binomial rv with pmf $nb(x; r, p)$, then

$$E(X) = \frac{r(1 - p)}{p} \quad \text{and} \quad V(X) = \frac{r(1 - p)}{p^2}$$

The Negative Binomial Distribution

Example 27. A pediatrician wishes to recruit 5 couples, each of whom is expecting their first child, to participate in a new natural childbirth regimen. Let

$$p = P(\text{a randomly selected couple agrees to participate})$$

If $p = .2$, what is the probability:

1. That 15 couples must be asked before 5 are found who agree to participate?
2. That at most 15 couples are observed before 5 are found who agree to participate?

The Negative Binomial Distribution

Example 27 (continued).

1. Let $S = \{\text{agrees to participate}\}$. We wish the probability that 10 F 's occur before the fifth S . Substituting $r = 5$, $p = .2$, and $x = 10$ into $nb(x; r, p)$ gives

$$nb(10; 5, .2) = \binom{14}{4} (.2)^5 (.8)^{10} = .034$$

2. The probability that at most 10 F 's are observed (at most 15 couples are asked) is

$$\begin{aligned} P(X \leq 10) &= \sum_{x=0}^{10} nb(x; 5, .2) \\ &= \sum_{x=0}^{10} \binom{x+4}{4} (.2)^5 (.8)^x \\ &= .164 \end{aligned}$$

The Negative Binomial Distribution

Remark

In the special case of $r = 1$, the pmf of X is

$$P(X = x) = nb(x; 1, p) = \binom{x}{0} p^1 (1 - p)^x = (1 - p)^x p$$

for $x = 0, 1, 2, \dots$. If Y is the number of trials necessary to obtain the first S (here $r = 1$, so we have only one S), then $Y = X + 1$. Hence, for $y = 1, 2, \dots$

$$P(Y = y) = P(X + 1 = y) = P(X = y - 1) = (1 - p)^{y-1} p$$

We refer to Y as **geometric random variable**, and its pmf is called the **geometric distribution**.

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The Poisson Probability Distribution

Definition

A discrete random variable X is said to have a **Poisson distribution** with parameter μ ($\mu > 0$) if X can take *any* non-negative integer value and has the following pmf:

$$p(x; \mu) = P(X = x) = e^{-\mu} \frac{\mu^x}{x!} \quad \text{for } x = 0, 1, 2, 3, \dots$$

The Poisson Probability Distribution

Proposition

The cdf of a Poisson rv X is denoted $F(x; \mu)$ and is given for *any* number x by

$$F(x; \mu) = \begin{cases} e^{-\mu} \sum_{y=0}^{\lfloor x \rfloor} \frac{\mu^y}{y!} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

The cdf of a Poisson distribution is tabulated for some values of μ in the Appendix.

The Poisson Probability Distribution

Example 28. Let X denote the number of defects in a particular type of metal oxide semiconductor transistor, and suppose it has a Poisson distribution with $\mu = 2$. Compute:

1. The probability that there are exactly three defects.
2. The probability that there are at most three defects.

The Poisson Probability Distribution

Example 28 (continued).

1. The probability that there are exactly three defects is

$$P(X = 3) = p(3; 2) = e^{-2} \frac{2^3}{3!} = .1804$$

2. The probability that there are at most three defects is

$$\begin{aligned} P(X \leq 3) &= F(3; 2) = \sum_{x=0}^3 p(x; 2) \\ &= e^{-2} \frac{2^0}{0!} + e^{-2} \frac{2^1}{1!} + e^{-2} \frac{2^2}{2!} + e^{-2} \frac{2^3}{3!} \\ &= e^{-2} \left(1 + 2 + 2 + \frac{8}{6} \right) = .8571 \end{aligned}$$

The latter cumulative probability can be found at the intersection of the $\mu = 2$ column and the $x = 3$ row of Appendix Table A.2. ■

The Poisson Probability Distribution

Proposition

If X has a Poisson distribution with parameter μ , then

$$E(X) = V(X) = \mu$$

Proof. These results can be derived directly from the definitions of mean and variance. For the mean,

$$\begin{aligned} E(X) &= \sum_{x=0}^{\infty} x \cdot p(x; \mu) = \sum_{x=1}^{\infty} x e^{-\mu} \frac{\mu^x}{x!} \\ &= e^{-\mu} \sum_{x=1}^{\infty} \frac{\mu^x}{(x-1)!} = e^{-\mu} \sum_{y=0}^{\infty} \frac{\mu^{y+1}}{y!} \\ &= e^{-\mu} \mu \sum_{y=0}^{\infty} \frac{\mu^y}{y!} = e^{-\mu} \mu e^{\mu} = \mu \end{aligned}$$

The Poisson Probability Distribution

Proof (continued). To compute the variance, we start by computing $E(X^2)$:

$$\begin{aligned} E(X^2) &= \sum_{x=0}^{\infty} x^2 \cdot p(x; \mu) = \sum_{x=1}^{\infty} x^2 e^{-\mu} \frac{\mu^x}{x!} = \sum_{x=1}^{\infty} x(x-1+1) e^{-\mu} \frac{\mu^x}{x!} \\ &= \sum_{x=1}^{\infty} x(x-1) e^{-\mu} \frac{\mu^x}{x!} + \sum_{x=1}^{\infty} x e^{-\mu} \frac{\mu^x}{x!} \\ &= e^{-\mu} \sum_{x=2}^{\infty} x(x-1) \frac{\mu^x}{x!} + \mu = e^{-\mu} \sum_{x=2}^{\infty} \frac{\mu^x}{(x-2)!} + \mu \\ &= e^{-\mu} \sum_{y=0}^{\infty} \frac{\mu^{y+2}}{y!} + \mu = e^{-\mu} \mu^2 \sum_{y=0}^{\infty} \frac{\mu^y}{y!} + \mu \\ &= e^{-\mu} \mu^2 e^{\mu} + \mu = \mu^2 + \mu \end{aligned}$$

Hence

$$V(X) = E(X^2) - [E(X)]^2 = \mu^2 + \mu - \mu^2 = \mu$$

The Poisson Probability Distribution

Proposition

Suppose that in the binomial pmf $b(x; n, p)$, we let $n \rightarrow \infty$ and $p \rightarrow 0$ in such a way that np approaches a value $\mu > 0$. Then

$$b(x; n, p) \rightarrow p(x; \mu)$$

According to this result, *in any binomial experiment in which n is large and p is small, $b(x; n, p) \approx p(x; \mu)$, where $\mu = np$. As a rule of thumb, this approximation can safely be applied if $n > 50$ and $np < 5$.*

The Poisson Probability Distribution

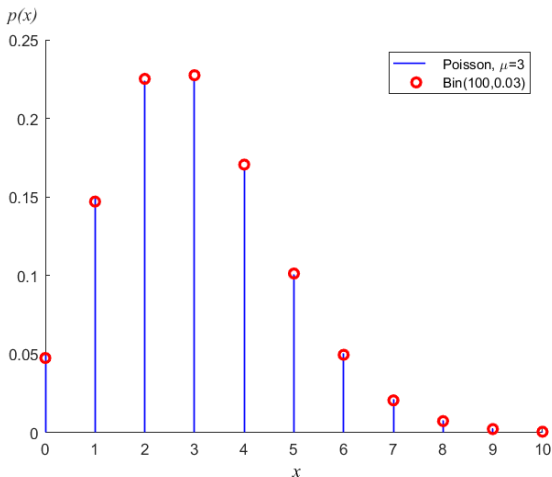
The following table shows the binomial distribution with $n = 100$ and $p = .03$ along with the Poisson distribution with $\mu = 3$, for $x = 0, 1, \dots, 10$.

x	Bin(100,0.03)	Poisson, $\mu = 3$
0	0.0476	0.0498
1	0.1471	0.1494
2	0.2252	0.2240
3	0.2275	0.2240
4	0.1706	0.1680
5	0.1013	0.1008
6	0.0496	0.0504
7	0.0206	0.0216
8	0.0074	0.0081
9	0.0023	0.0027
10	0.0007	0.0008

Clearly, that the accuracy of the approximation is good.

The Poisson Probability Distribution

We can plot the line graph of the Poisson distribution with $\mu = 3$ along with the binomial with $n = 100$ and $p = .03$, for $x = 0, 1, \dots, 10$.



The Poisson Probability Distribution

Example 29. A publisher of books takes great pains to ensure that its books are free of typographical errors, so that the probability of any given page containing at least one such error is .005 and errors are independent from page to page. What is the probability that one of its 600-page novels will contain exactly one page with errors? At most three pages with errors?

The Poisson Probability Distribution

Example 29 (continued). With S denoting a page containing at least one error and F an error-free page, the number X of pages containing at least one error is a binomial rv with $n = 600$ and $p = .005$.

1. We wish

$$\begin{aligned}P(X = 1) &= b(1; 600, .005) = \binom{600}{1} (.005)^1 (1 - .005)^{599} \\&= (600) \cdot (.995)^{599} = 0.1490\end{aligned}$$

Notice that since $n > 50$ and $np = 3 < 5$, we can approximate the binomial distribution with a Poisson distribution with parameter 3. This gives

$$p(1; 3) = e^{-3} \frac{3^1}{1!} = 3e^{-3} = 0.1494$$

Notice that the approximation is very good.

The Poisson Probability Distribution

Example 29 (continued).

2. Similarly, using the Poisson approximation,

$$P(X \leq 3) = F(3; 3) = \sum_{x=0}^3 p(x; 3) = .6472319$$

Here too, we could have computed the exact value of the probability which is

$$B(3; 600, .005) = .6472324$$

Both values are identical to five-decimal-place accuracy.



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The Poisson Process

A very important application of the Poisson distribution arises in connection with the occurrence of events of some type over time. Events of interest might be visits to a particular Web site, pulses of some sort recorded by a counter, email messages sent to a particular address, accidents in an industrial facility, or cosmic ray showers observed by astronomers at a particular observatory. We make the following assumptions about the way in which the events of interest occur:

1. There exists a parameter $\alpha > 0$ such that for any short time interval of length Δt , the probability that exactly one event occurs is $\alpha \cdot \Delta t + o(\Delta t)$.
2. The probability of more than one event occurring during Δt is $o(\Delta t)$.
3. The number of events occurring during the time interval Δt is independent of the number that occur prior to this time interval.

The Poisson Process

Definition

The occurrence of events over time as described is called a **Poisson process**; the parameter α specifies the **rate** for the process.

Proposition

In a Poisson process, the number of events occurring during a time interval of length t is a Poisson rv with parameter $\mu = \alpha t$, so that the probability that k events will be observed during any particular time interval of length t is

$$P_k(t) = e^{-\alpha t} \frac{(\alpha t)^k}{k!}$$

The expected number of events during any such time interval is then αt , so the expected number during a unit interval of time is α .

The Poisson Process

Example 30. Suppose pulses arrive at a counter at an average rate of six per minute, so that $\alpha = 6$. To find the probability that in a .5-min interval at least one pulse is received, note that the number of pulses in such an interval has a Poisson distribution with parameter $\alpha t = (6) \cdot (.5) = 3$. If we denote

X = the number of pulses received in the 30-sec interval

then

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-3} \frac{3^0}{0!} = 1 - e^{-3} = 0.9502$$



The Poisson Process

Instead of observing events over time, consider observing events of some type that occur in a two- or three-dimensional region. For example, we might select on a map a certain region R of a forest, go to that region, and count the number of trees. Each tree would represent an event occurring at a particular point in space. Under assumptions similar to 1, 2, and 3, it can be shown that the number of events occurring in a region R has a Poisson distribution with parameter $\alpha \cdot a(R)$, where $a(R)$ is the area of R . The quantity α is the expected number of events per unit area or volume.